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# BuildHub: A Hybrid Rule-Based and Machine Learning Framework for Pre-Construction Risk Forecasting in Residential Projects

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***Abstract*—Climate-sensitive construction projects are characterized by a higher likelihood of cost escalation and schedule postponement. These issues can be reduced when quantitative risk awareness is incorporated at the earliest phase of project planning. This paper presents a web-based decision-support system named BuildHub. The system generates quantitative risk awareness predictions through a hybrid combination of a deter- ministic domain rule engine and a machine learning classifier. A set of project parameters supplied by homeowners is transformed into domain characteristics such as monsoon exposure indices and district-level climate risk factors. These parameters are assessed by both the rule-based and machine learning pathways. Their outputs are combined through a confidence-sensitive blending layer. In addition, a hard-constraint override module is used to produce conservative forecasts when the input parameters fall at extreme values. A what-if recommendation mechanism and SHAP attributions are provided for each prediction instance to improve transparency. Two synthetically created datasets, each containing 4,000 instances, were generated and labeled according to residential construction patterns in Kerala. The machine learning component uses a Random Forest classifier. Under 5-fold stratified cross-validation, the model achieves a mean weighted F1-score of 0.9115 for cost overrun prediction and 0.9023 for schedule delay prediction. An ablation study shows that the machine learning pathway dominates in-distribution accuracy, while the hybrid model is expected to be more robust for non- training-distribution inputs.**

***Keywords*—*construction risk forecasting, cost overrun prediction, schedule delay, hybrid AI architecture, Random Forest, SHAP explainability, Kerala, pre-construction decision support***

## I. Literature Survey

Ensemble methods such as Gradient Boosting and Ran- dom Forest have demonstrated competitive performance in predicting cost and delay outcomes on tabular construction data [3]. Neural network techniques have also been applied to risk scheduling in large infrastructure projects, although their limited interpretability makes them less suitable for owner-facing applications [6]. A recurring observation across these studies is that feature engineering from raw project parameters often contributes more to predictive performance than classifier choice alone [3].Expert systems encoding heuristic construction knowledge have existed since the 1990s for tasks such as scheduling and procurement risk evaluation [4].

More recent studies have formalized these heuristics through fuzzy inference systems to produce graded risk values instead of binary decisions [7]. A key limitation of purely rule-based approaches is that they cannot capture statistical regularities not explicitly anticipated during system design.

Hybrid systems that integrate rule-based and learned components have been explored in areas such as medical diag- nosis, fraud detection, and industrial fault prediction [8]. A common finding is that hybrid models can improve robustness, particularly under uncertain or out-of-distribution conditions, whereas pure machine learning models often perform best on in-distribution test data. However, this paradigm has seen limited application in construction risk assessment.

SHAP values provide feature-level attributions for model predictions based on cooperative game theory and are applicable to tree-based models as well as other differentiable architectures [10]. Recent literature has reported the use of SHAP in construction safety and productivity-related prediction tasks [11]. Counterfactual explanations, in turn, identify the smallest plausible input changes required to alter a model's decision and can therefore provide meaningful decision support to users [12].

Synthetic data generation has been applied in construction research to address sparse project records by encoding do- main knowledge into simulated examples for cost modeling and schedule estimation [5]. A central validity concern is circularity: models trained on artificially labeled data may learn the labeling scheme rather than the underlying real- world phenomenon. This issue is explicitly acknowledged and examined in the evaluation design of the present work.

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## II. Introduction

Some of the most common reported failure modes in res- idential construction worldwide include budget overruns and schedule delays [1]. Studies in South and Southeast Asia have shown that a substantial proportion of owner-funded housing projects exceed original cost estimates, often by significant margins, and that time overruns are similarly common [2]. These tendencies are particularly relevant in the Indian state of Kerala due to strong monsoon seasonality, heterogeneous terrain, and an informal procurement environment in which homeowners often bear the financial risk without access to professional risk evaluation tools.

The pre-construction stage is the most suitable phase for quantitative risk prediction, since major contractual and pro- curement commitments have not yet been made. However, most prior research in construction risk modeling has focused primarily on commercial and infrastructure projects, where historical project data are more readily available for supervised learning [3]. Such approaches are less applicable in residential settings in emerging economies, where systematic longitudinal project records are often unavailable.

In the absence of deep historical data, two complementary strategies are relevant. Rule-based expert systems encode prac- titioner knowledge into deterministic scoring functions and do not require training data [4]. Supervised learning, in contrast, can be applied using synthetically generated data informed by domain knowledge when real records are scarce. However, neither strategy is fully satisfactory on its own: rule engines cannot capture unanticipated patterns, while models trained on synthetic data may fail to generalize beyond the distribution induced by the data generation process.

BuildHub addresses this gap through a hybrid architecture that combines both strategies. The rule engine provides a predictable and interpretable foundation, while the machine learning pathway captures non-linear feature interactions. A confidence- sensitive blending layer dynamically adjusts the relative contribution of each pathway during inference.

The main contributions of this work are as follows:

1) A hybrid pre-construction risk forecasting framework that combines a deterministic rule engine and a Random Forest classifier.
2) A Kerala-specific feature engineering pipeline incor- porating monsoon exposure, terrain, and flood-related district risk.
3) A confidence-sensitive blending mechanism for combin- ing rule-based and machine learning predictions.
4) A hard-constraint safety override module for conserva- tive prediction under extreme input conditions.
5) Integration of SHAP-based per-instance explanations and counterfactual-style recommendations for trans- parency.
6) A complete web-based decision-support system de- signed for homeowner-facing use without requiring tech- nical expertise.

## III. Methodology

Two independent datasets were created through a deter- ministic labeling process: one for cost overrun risk and one for schedule delay risk. Each dataset consists of $N = 4000$ records, generated using a random seed of 42. Input features were sampled from distributions aligned with Kerala residen- tial construction norms. Building footprints ranged from 500 to 5,000 square feet, plot sizes ranged from 3 to 20 cents, project budgets were distributed across five stratified categories, and planned durations ranged from 6 to 36 months.

Each synthetic record was assigned to one of the fourteen districts of Kerala. The climate profile of each district is rep- resented through three ordinal features: terrain code, rainfall code, and flood risk code. These profiles were informed by district-level rainfall and flood hazard characteristics.

Risk labels were assigned using deterministic domain rules based on the interaction among budget adequacy, project duration, monsoon exposure, design complexity, and district- level climate risk. The labeling procedure generated three classes for each task: Low, Medium, and High.

Raw user inputs were transformed into engineered features before model training. "Budget per sqft" captures the adequacy of the budget relative to total floor area. "Design complexity score" models the interaction between number of floors and total area. "Monsoon exposure score" estimates the proportion of the planned project timeline that overlaps with the monsoon season. "Effective monsoon score" weights monsoon exposure by district rainfall severity.

Processed homeowner input data pass through multiple sequential stages, from feature engineering and risk inference to explanation and recommendation generation, as shown in Fig. 1.

The rule engine is based on threshold-driven heuristics that compute three independent sub-scores: cost score, time score, and complexity score. These are normalized to the unit interval and mapped to risk levels using fixed thresholds.

Each prediction task is modeled using a Random Forest classifier with 300 estimators, maximum tree depth of 10, and balanced class weighting. Although Gradient Boosting achieved slightly better cross-validation performance, Random Forest was selected because it supports efficient per- instance explanation using SHAP TreeExplainer [10], [13], [14].

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The blended risk score is computed as:

$$\text{Blended Score} = w_{ML} \cdot s_{ML} + (1 - w_{ML}) \cdot s_{rule} \quad (1)$$

where $w_{ML} = 0.40$ when the classifier's maximum class probability exceeds 0.55, and $w_{ML} = 0.15$ otherwise.

A hard-constraint override mechanism is used to ensure conservative prediction in extreme scenarios. To improve transparency, each prediction is accompanied by SHAP-based feature attributions and a what-if recommendation mechanism.

## IV. Results and Discussion

The experiments were conducted using an 80/20 stratified train-test split with a random seed of 42, along with five-fold stratified cross-validation. Three classifiers were considered: Logistic Regression (LR), Random Forest (RF), and Gradient Boosting (GB). The evaluation metrics included weighted F1-score and class-specific F1-scores.

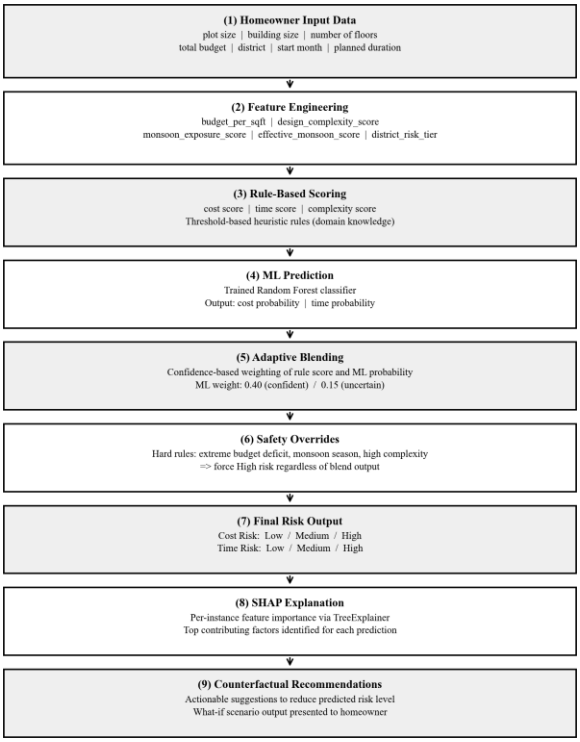



*Fig. 1. BuildHub hybrid risk prediction workflow*

*TABLE I*
*Baseline Model Comparison (80/20 Stratified Split)*

| Task | Model | Acc. | W-F1 | F1Low | F1High |
|---|---|---|---|---|---|
| Cost | LR | 0.8712 | 0.8765 | 0.0000 | 0.9203 |
| Cost | RF | 0.9012 | 0.8970 | 0.3077 | 0.9378 |
| Cost | GB | 0.9225 | 0.9204 | 0.3636 | 0.9527 |
| Time | LR | 0.9113 | 0.9134 | 0.5455 | 0.9495 |
| Time | RF | 0.9050 | 0.9042 | 0.6957 | 0.9417 |
| Time | GB | 0.9087 | 0.9061 | 0.6667 | 0.9509 |

*TABLE II*
*Five-Fold Stratified Cross-Validation (Weighted F1)*

| Task | Model | Mean F1 | Std |
|---|---|---|---|
| Cost | RF | 0.9115 | 0.0085 |
| Cost | GB | 0.9329 | 0.0080 |
| Cost | LR | 0.8821 | 0.0091 |
| Time | RF | 0.9023 | 0.0117 |
| Time | GB | 0.9213 | 0.0123 |
| Time | LR | 0.8937 | 0.0075 |

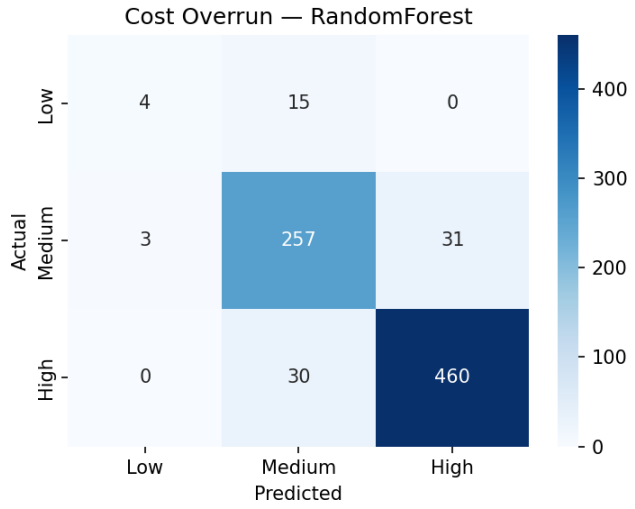



*Fig. 2. Confusion matrix — Cost Overrun Risk (Random Forest, held-out test set)*

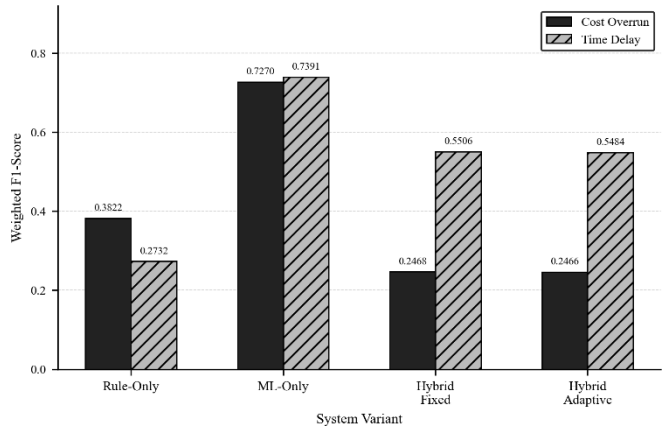



*Fig. 3. Comparison of rule-based, machine learning, and hybrid variants of prediction based on weighted F1-score through ablation study*

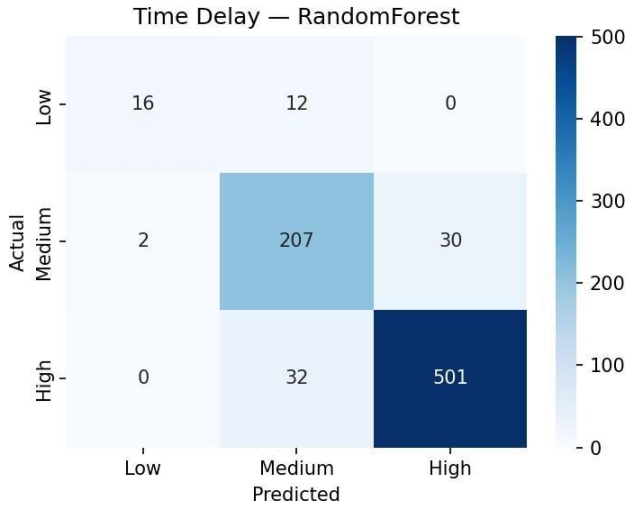



*Fig. 4. Confusion matrix — Time Delay Risk (Random Forest, held-out test set).*

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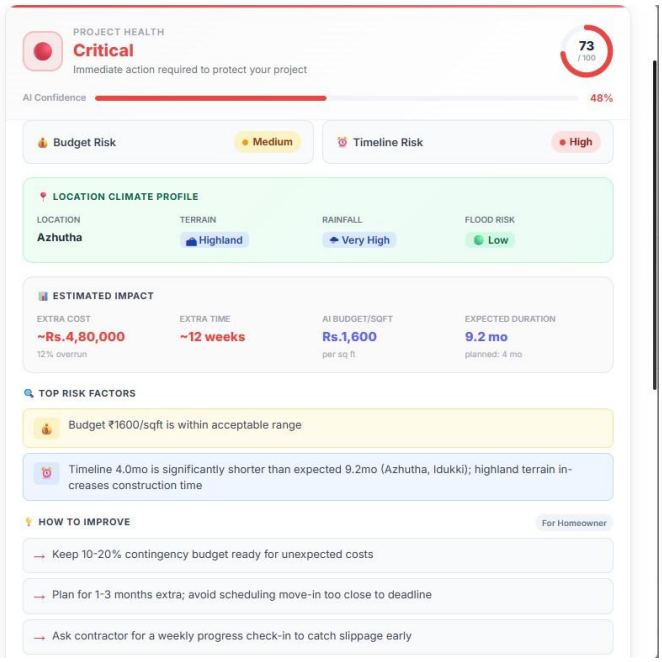



*Fig. 5. BuildHub risk assessment panel displaying predicted risk levels, SHAP feature attributions, and counterfactual recommendations.*

Table I reports the baseline model comparison, while Table II presents the five-fold cross-validation results. Gradient Boosting outperformed Random Forest on both tasks. However, because the margin was modest and Random Forest provides superior compatibility with SHAP-based local explanation, RF was selected as the deployment model.

Figure 2 compares four variants: rule-only, ML-only, hybrid with fixed blend weights, and hybrid with adaptive blend weights. The ML-only variant achieved the best weighted F1-score on both tasks on the synthetic in-distribution test set. This outcome is expected due to the circular nature of the data generation process.

Figures 3 and 4 present the confusion matrices of the deployed Random Forest model on the held-out test partition. The most common error in both tasks is confusion between Medium and High risk classes.

BuildHub is implemented as a three-layer web application. The presentation layer is built with React.js, the application layer uses PHP with a relational database, and the inference layer is a FastAPI-based microservice containing the Random Forest models, rule engine, blending logic, safety override module, and SHAP computation.

Figure 5 shows the risk assessment panel presented to homeowners after project parameter submission. The panel displays predicted cost overrun and schedule delay risk levels, the most influential SHAP features, and counterfactual recommendations.

A key limitation of the study is that both datasets are fully synthetic. Since the classifier is trained on labels produced by a deterministic rule procedure, measured performance largely reflects agreement with that labeling mechanism rather than direct prediction of actual construction outcomes. Therefore, although the hybrid architecture is conceptually motivated for robustness, real-world validation remains necessary.

## V.CONCLUSION

This paper introduced BuildHub, a hybrid pre-construction risk forecasting system that combines a deterministic rule engine with a trained Random Forest classifier through adap- tive confidence-weighted blending. The system provides three- class prediction of cost overrun and schedule delay risk, along with SHAP-based explanations and counterfactual-style recommendations, in a homeowner-facing web application.

Experimental analysis on synthetically generated Kerala-responsive datasets showed strong cross-validation performance and confirmed the usefulness of balanced class weighting for minority-class handling. The ablation study demonstrated that the machine learning module dominates in-distribution accuracy, while the hybrid design is expected to improve robustness when the classifier encounters uncertain or out-of-distribution inputs.

The primary limitation of the current work is the absence of real-world project outcome data for model training and validation. Future work should focus on collecting completed residential project records from Kerala, incorporating contrac- tor performance history and real-time material price indices as additional features, and extending the framework toward dynamic in-construction risk monitoring.

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